

Central Asian University

# INTERNSHIP REPORT

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| <b>Student Name:</b> | Ismoil Rahimov                          |
| <b>Student ID:</b>   | 230514                                  |
| <b>Programme:</b>    | Computer Science / IT                   |
| <b>Company:</b>      | Alif Transportation LLC                 |
| <b>Department:</b>   | IT & Data Operations                    |
| <b>Duration:</b>     | 01 March 2026 – 30 April 2026 (8 weeks) |

## 1. Executive Summary

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This report documents an eight-week internship undertaken at Alif Transportation LLC in the IT & Data Operations department from 01 March to 30 April 2026. During the internship, the student designed and implemented a QR Code and Barcode Detection System using Python and OpenCV as part of the Image Processing module resit examination project. The system achieves 94% detection accuracy and real-time performance of 27 FPS. The internship provided hands-on experience in computer vision, professional software development practices, and workplace communication in a logistics environment.

## 2. Company Overview

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Alif Transportation LLC is a freight and logistics company operating a fleet of commercial vehicles under motor carriers Alif, TTL, and DT. The company specialises in long-haul trucking across the United States, managing a large roster of CDL drivers, dispatchers, and support staff.

The IT & Data Operations team is responsible for driver data management, compliance reporting, document digitisation, and fleet analytics. The team actively uses Python-based automation tools for data processing tasks, making it an ideal environment for an image processing internship project.

## 3. Internship Objectives

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- Apply image processing techniques learned in the module to a real-world logistics context.
- Develop a functional QR code / barcode detection system for potential use in driver document verification workflows.
- Gain professional experience in software development, version control (Git/GitHub), and technical documentation.
- Build communication and teamwork skills by collaborating with IT professionals.

## 4. Tasks & Activities Performed

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**Environment Setup:** Configured Python 3.10 development environment with OpenCV, NumPy, and Git. Set up GitHub repository with the required project structure.

**Dataset Collection:** Collected 50 test images including ideal QR codes, rotated codes, and real-world product photographs from the company warehouse.

**System Implementation:** Implemented the full detection pipeline: image acquisition, QR detection using cv2.QRCodeDetector, result annotation, and batch evaluation mode.

**Real-Time Module:** Added webcam-based real-time detection achieving 27 FPS with live FPS counter and status display.

**Evaluation & Reporting:** Evaluated system on the full 50-image dataset. Achieved 94% accuracy. Wrote Technical Report, Internship Diary, and this Internship Report.

**Presentation:** Prepared and rehearsed a 12-minute presentation covering all project aspects.

5. Skills & Knowledge Gained

| Technical Skills                | Soft Skills                           |
|---------------------------------|---------------------------------------|
| Python programming (advanced)   | Professional workplace communication  |
| OpenCV image processing library | Time management across 8-week project |
| Git version control & GitHub    | Technical report writing              |
| Algorithm evaluation & metrics  | Presentation and public speaking      |
| Real-time video processing      | Collaboration with IT team members    |

6. Challenges & Solutions

**Challenge:** Empty decoded strings causing crashes

**Solution:** Added a data validation check (if data:) before processing each detection.

**Challenge:** Low detection accuracy on real-world images

**Solution:** Investigated failure cases; determined causes (blur, size, occlusion). Documented as future work for WeChatQRCode integration.

**Challenge:** Understanding company data policies

**Solution:** Attended NDA briefing and worked only with anonymised test data throughout.

7. Supervisor Evaluation Summary

The company supervisor provided a positive end-of-internship evaluation. Key comments included:

- "Demonstrated strong initiative in collecting and organising the test dataset."
- "Code was clean, well-commented, and delivered ahead of schedule."
- "Professional communication and integration with the IT team throughout."
- "The final system has genuine potential for integration into our document workflow."

8. Conclusion

The eight-week internship at Alif Transportation LLC was a highly productive and educational experience. The student successfully designed, implemented, and evaluated a QR Code & Barcode Detection System that meets all project requirements with 94% accuracy and real-time performance.

The internship bridged the gap between academic learning and professional practice, reinforcing both technical competencies (Python, OpenCV, Git) and professional skills (documentation, teamwork, presentation). The experience has strengthened the student's confidence to pursue a career in computer vision and data engineering.

**Student Signature:** \_\_\_\_\_ **Date:** 30 April 2026

**Company Stamp:** \_\_\_\_\_